

LECTURE 9: UNIFORMLY MOST POWERFUL TESTS

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REFERENCE CB, Chapter 8.2.1 and 8.3.2, HMC, Chapter 8.1–2, H22, Chapter 13.10–12

1. NEYMAN-PEARSON LEMMA

Consider testing the null $H_0: \theta \in \Theta_0$ against the alternative $H_1: \theta \in \Theta_1$ based on data $X \sim F(\cdot \mid \theta)$. Recall that a maximin test maximizes the minimum power under the alternative subject to the level constraint

$$\sup_{\theta \in \Theta_0} E_{\theta}[\phi(X)] \leq \alpha.$$

In general, finding a maximin test is hard, because we need to optimize over the space of functions ϕ . If both the null and alternative are simple, then this problem turns out to have an elegant solution, due to Neyman and Pearson (1933), published in the world's oldest scientific journal (established in 1665). Egon Pearson was Karl Pearson's son.

To state the result, we need to allow for randomized decisions via *randomized tests*, so $\phi(x) \in [0, 1]$. The value $\phi(x)$ represents the probability of rejecting the null when we observe $X = x$. When $0 < \phi(x) < 1$, the decision is randomized. Randomized tests are seldom used in practice, but they are a useful theoretical device.

Lemma 1 (Neyman-Pearson, HMC Theorem 8.1.1, CB Theorem 8.3.12). Let P_0 and P_1 be distributions with densities f_0 and f_1 . Consider testing $H_0: X \sim P_0$ against the alternative $H_1: X \sim P_1$. Then the test

$$\phi(x) = \begin{cases} 1 & \text{if } f_1(x) > \lambda f_0(x), \\ \kappa & \text{if } f_1(x) = \lambda f_0(x), \\ 0 & \text{if } f_1(x) < \lambda f_0(x), \end{cases} \quad (1)$$

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where $\lambda \geq 0$ and $\kappa \in [0, 1]$ are chosen to satisfy $E_{P_0}[\phi(X)] = \alpha$, is most powerful among all tests with level α .

Proof. Let $\tilde{\phi}(x)$ be any other test with level α . We need to show that

$$\int \tilde{\phi}(x) f_1(x) dx \leq \int \phi(x) f_1(x) dx.$$

Note that, by definition of $\phi(x)$,

$$0 \leq (\phi(x) - \tilde{\phi}(x))(f_1(x) - \lambda f_0(x)).$$

Integrating both sides with respect to x and using the fact that $\tilde{\phi}$ has null rejection rate equal to at most α then yields

$$\begin{aligned} 0 &\leq \int (\phi(x) - \tilde{\phi}(x))(f_1(x) - \lambda f_0(x)) dx \\ &= \int \phi(x) f_1(x) dx - \lambda \int \phi(x) f_0(x) dx - \int \tilde{\phi}(x) f_1(x) dx + \lambda \int \tilde{\phi}(x) f_0(x) dx \\ &\leq \int \phi(x) f_1(x) dx - \lambda \alpha - \int \tilde{\phi}(x) f_1(x) dx + \lambda \alpha = \int \phi(x) f_1(x) dx - \int \tilde{\phi}(x) f_1(x) dx, \end{aligned}$$

which proves the result. $\square$

Remark 1. From this proof, it also follows that any other test $\tilde{\phi}$ that is also most powerful has to satisfy (i) $\tilde{\phi}(x) = 1$ if $f_1(x) > \lambda f_0(x)$, and $\tilde{\phi}(x) = 0$ if $f_1(x) < \lambda f_0(x)$ for almost all x , and (ii) $E_{P_0}[\tilde{\phi}(X)] = \alpha$, unless we are in a degenerate case with $\lambda = 0$. To see this, if $\tilde{\phi}$ is also most powerful, then it follows from the proof that we must have $(\phi(x) - \tilde{\phi}(x))(f_1(x) - \lambda f_0(x)) = 0$ for almost all x , and hence that $\phi(x) = \tilde{\phi}(x)$ for almost all x such that $f_1(x) \neq \lambda f_0(x)$. The only degree of freedom is what to do if $f_1(x) = \lambda f_0(x)$.

Furthermore, it also follows that any test that has the form in eq. (1), for some κ and λ , must be most powerful at level $\tilde{\alpha} = E_{P_0}[\phi(X)]$.

- The function $\text{LR}(x) = f_1(x)/f_0(x)$ is called the *likelihood ratio*. The lemma says that the most powerful test rejects for large values of the likelihood ratio (if the support of X under the null is smaller, then this is still true if we define $\text{LR}(x) = \infty$ for $f_0(x) = 0$).
- The randomization device ensures that we can make the size equal to α exactly. It's not necessary when the likelihood ratio is continuously distributed under the null.

Example 1. Let $X_1, \dots, X_n$ be a random sample from a $\mathcal{N}(\theta, \sigma^2)$ distribution with known σ^2 . Suppose we want to test $H_0: \theta = \theta_0$ against $H_1: \theta = \theta_1$. The likelihood ratio is

$$\begin{aligned} \frac{\prod_{i=1}^n \exp(-\frac{1}{2\sigma^2}(X_i - \theta_1)^2)}{\prod_{i=1}^n \exp(-\frac{1}{2\sigma^2}(X_i - \theta_0)^2)} &= \exp\left(-\frac{1}{2\sigma^2} \left(\sum_i (X_i - \theta_1)^2 - \sum_i (X_i - \theta_0)^2\right)\right) \\ &\propto \exp\left(\frac{n}{\sigma^2} \bar{X}_n(\theta_1 - \theta_0)\right), \end{aligned}$$

where $\bar{X}_n$ is the sample mean. By the Neyman-Pearson lemma, the most powerful test therefore rejects for large values of $\bar{X}_n$ if $\theta_1 > \theta_0$, and small values of $\bar{X}_n$ if $\theta_1 < \theta_0$. Consider the first case. Then the most powerful test rejects whenever $\bar{X}_n \geq c$, where the critical value is chosen to satisfy the level constraint, $P_{\theta_0}(\bar{X}_n \geq c) = \alpha$, which gives $c = \theta_0 + \sigma z_{1-\alpha} / \sqrt{n}$, where $z_{1-\alpha}$ is the $1 - \alpha$ quantile of the standard normal distribution. So the usual one-sided z-test is most powerful. $\square$

Example 2. Let X be a scalar with distribution $f_0(x) = \mathbb{1}\{0 \leq x \leq 1\}$ under the null, and $f_1(x) = 2x \mathbb{1}\{0 \leq x \leq 1\}$ under the alternative. Here the likelihood ratio is $2x$ on the support of X , so that the test rejects for large values of X . Since X is uniform under the null, the appropriate critical value is $1 - \alpha$, which yields $\phi(X) = \mathbb{1}\{X \geq 1 - \alpha\}$. $\square$

Example 3. We wish to test the null $H_0: \theta = \theta_0$ against the alternative $H_1: \theta = \theta_1$ for some $\theta_1 > \theta_0$, based on observing n draws from a Bernoulli distribution, $X_i \sim \text{Bern}(\theta)$. Let $S = \sum_{i=1}^n X_i$ denote the sum. The joint likelihood is $\theta^S (1 - \theta)^{n-S}$, so that

$$\text{LR}(X_1, \dots, X_n) = \left(\frac{\theta_1}{\theta_0}\right)^S \left(\frac{1 - \theta_1}{1 - \theta_0}\right)^{n-S} = \left(\frac{\theta_1 / (1 - \theta_1)}{\theta_0 / (1 - \theta_0)}\right)^S \left(\frac{1 - \theta_1}{1 - \theta_0}\right)^n,$$

which is increasing in S . So we reject for large values of S . To satisfy the level constraint, we will generally need to randomize. For example, if $\theta_0 = 1/2$, $n = 2$, and $\alpha \leq 1/4$, we reject with probability $\kappa = 4\alpha$ if $S = 2$, and don't reject otherwise. $\square$

2. COMPOSITE HYPOTHESES

The Neyman-Pearson lemma solves the problem of testing a simple null against a simple alternative. This is usually not that interesting in practice. However, in select scenarios, the lemma can be extended to find uniformly most powerful (UMP) tests: that is, tests that, subject to the level constraint, maximize the power $E_\theta[\phi(X)]$ *simultaneously* for all $\theta \in \Theta_1$. Here we discuss one such scenario. First a definition:

Definition 1. A family of densities $f(x | \theta)$ with $\theta \in \mathbb{R}$ has a *monotone likelihood ratio* if there exists some function $T(x)$ such that for any $\theta < \theta'$, the ratio $f(x | \theta') / f(x | \theta)$ is a non-decreasing function of $T(x)$.

Theorem 2 (The Karlin–Rubin theorem, HMC, p. 443, CB, Theorem 8.3.17). Let $f(x | \theta)$, $\theta \in \mathbb{R}$ be a parametric family with monotone likelihood ratio in $T(x)$. Suppose that the family is identifiable, so that for all $\tilde{\theta} \neq \theta$, $P_\theta(\{x: f(x | \theta) \neq f(x | \tilde{\theta})\}) > 0$. Consider testing the null $H_0: \theta \leq \theta_0$ against the alternative $H_1: \theta > \theta_0$ at level α based on an observation X with density $f(\cdot | \theta)$. Then the test

$$\phi(x) = \begin{cases} 1 & \text{if } T(x) > \lambda, \\ \kappa & \text{if } T(x) = \lambda, \\ 0 & \text{if } T(x) < \lambda, \end{cases}$$

where $\kappa \in [0, 1]$ and $\lambda \in \mathbb{R}$ are chosen to satisfy $E_{\theta_0}[\phi(X)] = \alpha$, is UMP. In addition, the power of this test $\pi(\theta) = E_{\theta}[\phi(X)]$ is strictly increasing in θ for all θ such that $\pi(\theta) \in (0, 1)$.

Proof. Fix some simple alternative $\theta_1 > \theta_0$. By the Neyman-Pearson lemma, the most powerful test of the null $\theta = \theta_0$ against the alternative $\theta = \theta_1$ rejects for large values of $f(X | \theta_1)/f(X | \theta_0)$. By the monotone likelihood ratio property, this ratio depends on X through $T(X)$, and is monotone in $T(X)$. Therefore, the test rejects for large values of $T(X)$, which does not depend on the particular alternative θ_1 .

To show that the power function is increasing in θ , observe that by Remark 1, there exists a level $\tilde{\alpha}$ such that this test is also most powerful for $H_0: \theta = \tilde{\theta}_0$ against $H_1: \theta = \tilde{\theta}_1$, for any $\tilde{\theta}_0 < \tilde{\theta}_1$. It also must be the case that $\tilde{\alpha} = \pi(\tilde{\theta}_0) \leq \pi(\tilde{\theta}_1)$, since the test must be more powerful than the test $\tilde{\phi}(x) = \tilde{\alpha}$. Furthermore, by the identifiability condition, the inequality must be strict. Otherwise $\tilde{\phi}(x)$ would also be most powerful, which by Remark 1 implies that $f(x | \tilde{\theta}_0) = f(x | \tilde{\theta}_1)$ for almost all x , contradicting identifiability. So it must be that the power function is strictly increasing, from which it also follows that the test controls size. $\square$

Example 3 (continued). The binomial distribution also has a monotone likelihood ratio, so tests of the hypothesis $H_0: \theta \leq \theta_0$ against $H_1: \theta > \theta_0$ reject for large values of S . $\square$

Example 1 (continued). This distribution has a monotone likelihood ratio, so the (one-sided) z-test is UMP for testing $H_0: \theta \leq \theta_0$ against $H_1: \theta > \theta_0$. $\square$

The following generalization of this result is one of the most important results in statistics.

Theorem 3. Let $X \sim \mathcal{N}(\theta, \Sigma)$, with the covariance matrix Σ known, and consider testing the null $H_0: \theta_1 \leq \vartheta$, $(\theta_2, \dots, \theta_k) \in \mathbb{R}^{k-1}$ against the alternative $H_1: \theta_1 > \vartheta$, $(\theta_2, \dots, \theta_k) \in \mathbb{R}^{k-1}$, where ϑ is some constant. Then the one-sided z-test based on X_1 , $\phi(X) = \mathbb{1}\{X_1 - \vartheta \geq z_{1-\alpha} \Sigma_{11}^{1/2}\}$ is UMP.

You will prove this theorem in the homework. Why is it so important? Two reasons. First, consider the linear regression model $Y = X\beta + u$, and suppose $u \sim \mathcal{N}(0, \sigma^2 I_n)$, with σ^2 known. Then, treating the regressors as fixed, the least squares estimator $\hat{\beta} = (X'X)^{-1}X'Y$ is a sufficient statistic. But $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2(X'X)^{-1})$. So for testing one-sided nulls about the regression parameters, the one-sided z-test is UMP. This is why we use z-tests in linear regression analysis. Furthermore, it follows that one cannot achieve power gains over the z-test by fancier methods, such as model selection, the lasso, elastic net, etc. Such methods must either fail to control size, or else they must lead to power loss. The second reason will be given in the next lecture, when we discuss likelihood-based tests. When you are choosing which single result from this course to remember, please choose this one!

3. UNBIASEDNESS

In most cases, no UMP test exists. This is the case whenever the alternative is two-sided.

Example 4. Let $X_1, \dots, X_n$ be a random sample from a $\mathcal{N}(\theta, \sigma^2)$ distribution with known σ^2 . Suppose we want to test $H_0: \theta = \theta_0$ against the alternative hypothesis, $H_1: \theta \neq \theta_0$. We know that the UMP test against alternatives that are bigger than θ_0 rejects for large values of $\bar{X}_n$, but for alternatives smaller than θ_0 , we'd want to reject for small values of $\bar{X}_n$. So no test can simultaneously be UMP against alternatives Θ_1 that contain values both bigger and smaller than θ_0 . $\square$

Amazingly, in some special cases, UMP tests against two-sided alternatives exist once we impose unbiasedness.

Definition 2. A level α test ϕ is *unbiased* if its power function satisfies $\pi(\theta) \geq \alpha$ for all $\theta \in \Theta_1$.

While unbiasedness can be problematic for estimation, for testing it makes sense. For one thing, maximin tests are always unbiased (why?). Furthermore, similar to the one-sided case, if a maximin test against alternatives of the form $H_1: |\theta - \theta_0| \geq d$ does not depend on the choice of d , then it is UMP unbiased. The most important example where this holds is the normal means model, illustrated next.

Example 4 (continued). By a sufficiency argument, it is enough to consider tests based on $\bar{X}_n \sim \mathcal{N}(\theta, \sigma^2/n)$. To save on notation, write X instead of $\bar{X}_n$, and normalize its variance to 1. So the task is to find the UMP unbiased test based on a single observation $X \sim \mathcal{N}(\theta, 1)$.

Consider some test ϕ . Its power is

$$\pi(\theta) = \int_{\mathbb{R}} \phi(x) f(x - \theta) dx,$$

where $f(x) = (2\pi)^{-1/2} e^{-\frac{x^2}{2}}$ denotes the probability density function (PDF) of a standard normal distribution. Differentiating under the integral sign using the chain rule and the fact that $f'(x) = -xf(x)$ yields

$$\pi'(\theta) = \int_{\mathbb{R}} \phi(x) f(x - \theta)(x - \theta) dx.$$

If ϕ is unbiased, then we necessarily must have $\pi'(\theta_0) = 0$, since $\pi(\theta)$ is minimized at θ_0 . Fix some $\theta_1 \neq \theta_0$. The most powerful unbiased test against this alternative therefore maximizes

$$\int_{\mathbb{R}} \phi(x) f(x - \theta_1) dx.$$

subject to

$$\int_{\mathbb{R}} \phi(x) f(x - \theta_0)(x - \theta_0) dx = 0, \quad \int_{\mathbb{R}} \phi(x) f(x - \theta_0) dx = \alpha. \quad (2)$$

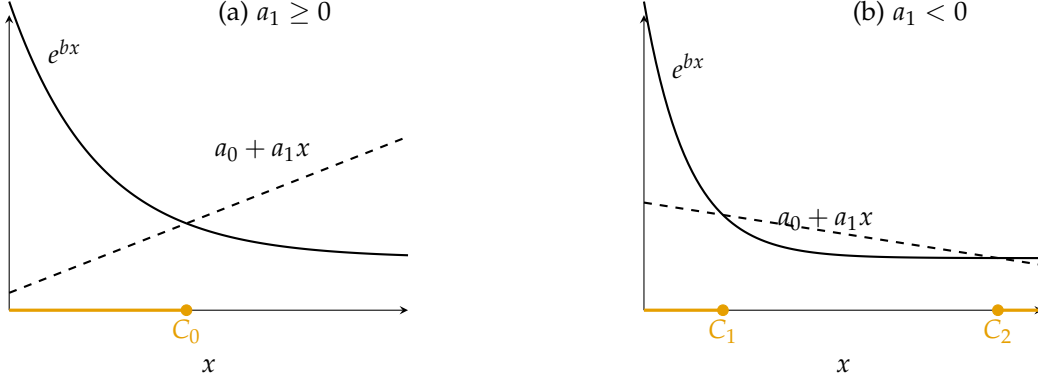


Figure 1: Two possible shapes of the rejection region, highlighted in orange, for the test in eq. (3) when $b < 0$, depending on whether a_1 is positive.

We now generalize the Neyman-Pearson lemma to claim that a test of the form

$$\phi(x) = \begin{cases} 1 & \text{if } f(x - \theta_1) > \lambda_1 f(x - \theta_0) + \lambda_2 f(x - \theta_0)(x - \theta_0), \\ 0 & \text{if } f(x - \theta_1) < \lambda_1 f(x - \theta_0) + \lambda_2 f(x - \theta_0)(x - \theta_0). \end{cases} \quad (3)$$

for some λ_1, λ_2 chosen to satisfy the constraints (2) solves this problem (this is like the Neyman-Pearson test, except the “null density” is $\lambda_1 f(x - \theta_0) + \lambda_2 f(x - \theta_0)(x - \theta_0)$). We don’t need to consider randomized tests since the distribution of the new “likelihood ratio” is continuous. To prove the claim, consider any other unbiased test $\tilde{\phi}$. Then

$$(\phi - \tilde{\phi})(f(x - \theta_1) - \lambda_1 f(x - \theta_0) - \lambda_2 f(x - \theta_0)(x - \theta_0)) \geq 0,$$

so integrating with respect to x yields

$$\begin{aligned} 0 &\leq E_{\theta_1}[\phi - \tilde{\phi}] - \lambda_1 E_{\theta_0}[\phi - \tilde{\phi}] - \lambda_2 \int (\phi(x) - \tilde{\phi}(x)) f(x - \theta_0)(x - \theta_0) dx \\ &= E_{\theta_1}[\phi - \tilde{\phi}] - \lambda_1 E_{\theta_0}[\phi - \tilde{\phi}] = E_{\theta_1}[\phi - \tilde{\phi}], \end{aligned}$$

where the second line follows since the tests are unbiased, and the last equality uses the fact that since the power function is continuous, unbiasedness implies both tests have size α . This proves the claim that ϕ maximizes power at θ_1 among all unbiased tests.

Next, we show that the test ϕ does not depend on the choice of θ_1 . Observe that ϕ rejects if $e^{-(x-\theta_1)^2/2} > \lambda_1 e^{-(x-\theta_0)^2/2} + \lambda_2 e^{-(x-\theta_0)^2/2}(x - \theta_0)$, or, equivalently, if

$$e^{xb} > a_0 + a_1 x,$$

where $b = \theta_1 - \theta_0$, $a_0 = e^{\theta_1^2/2 - \theta_0^2/2}(\lambda_1 - \lambda_2 \theta_0)$, and $a_1 = e^{\theta_1^2/2 - \theta_0^2/2} \lambda_2$.

There are two cases to consider: $b < 0$ or $b > 0$, depending on whether we fixed θ_1 to be greater or smaller than θ_0 . If $b < 0$, and $a_1 \geq 0$, then the rejection region is one-sided

(see panel (a) of Figure 1), which cannot be the case since a one-sided rejection region has a strictly non-zero power derivative at θ_0 , violating the constraint $\pi'(\theta_0) = 0$ established above. So if $b < 0$, we must have $a_1 < 0$, and the rejection region is a complement of an interval (see panel (b) of Figure 1), $\phi(X) = \mathbb{1}\{X \notin [C_1, C_2]\}$.

By a symmetric argument, if $b > 0$, we must have $a_1 > 0$, and the rejection region has the same shape. But to satisfy the unbiasedness requirement, it must be the case that the midpoint of the interval $[C_1, C_2]$ is θ_0 . To see this, note that $\int (x - \theta_0)f(x - \theta_0) dx = E_{\theta_0}[X - \theta_0] = 0$, so that $\pi'(\theta_0) = \int \mathbb{1}\{x \notin [C_1, C_2]\}(x - \theta_0)f(x - \theta_0) dx = - \int_{C_1}^{C_2} (x - \theta_0)f(x - \theta_0) dx = - \int_{C_1 - \theta_0}^{C_2 - \theta_0} xf(x) dx$. This equals zero, by a symmetry argument, only if $C_2 - \theta_0 = -(C_1 - \theta_0) \iff (C_1 + C_2)/2 = \theta_0$. So the acceptance region is $[2\theta_0 - C_2, C_2]$. The size constraint gives $C_2 - \theta_0 = z_{1-\alpha/2}$, the $1 - \alpha/2$ quantile of the standard normal distribution. Therefore, the UMP unbiased test rejects when $|X - \theta_0| \geq z_{1-\alpha/2}$. Since this test doesn't depend on the particular alternative θ_1 , it must be UMP unbiased against $H_1: \theta \neq \theta_0$.

Translating this to the original setup, we conclude that the two-sided z-test that rejects whenever $|\bar{X}_n - \theta_0| \geq z_{1-\alpha/2}\sigma/\sqrt{n}$ is UMP unbiased for testing $H_0: \theta = \theta_0$ against a two-sided alternative. $\square$

Example 4 provides the most important example of a UMP unbiased test because the example generalizes along the lines of Theorem 3. As a result, the two-sided z-test in linear regression is UMP unbiased.

REFERENCES

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